

3.2.

$$\langle f, g \rangle = \int_1^5 f(x)g(x)dx$$

a)

$$\text{span}\{1, x, x^2\}$$

→ označme gúzanu ortogonálnu bázu $B = (\vec{u}_1, \vec{u}_2, \vec{u}_3)$

$$\vec{u}_1 = \frac{1}{\|1\|} = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{2}$$

$$\vec{u}_2 = \left(\int_1^5 \frac{x}{2} dx \right) \cdot \frac{1}{2} = 3$$

$$\vec{u}_2 = \frac{x - \vec{u}_2}{\|x - \vec{u}_2\|} = \frac{x - 3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}(x-3)}{4}$$

$$\vec{u}_3 = \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 dx \right) \cdot \frac{\sqrt{3}(x-3)}{4} + \left(\int_1^5 \frac{x^2}{2} dx \right) \cdot \frac{1}{2}$$

$$= 6(x-3) + \frac{31}{3}$$

$$\vec{u}_3 = \frac{x^2 - \vec{u}_3}{\|x^2 - \vec{u}_3\|} = \frac{x^2 - 6(x-3) - \frac{31}{3}}{\sqrt{\int_1^5 \left(x^2 - 6(x-3) - \frac{31}{3}\right)^2 dx}} = \frac{x^2 - 6(x-3) - \frac{31}{3}}{\sqrt{\frac{256}{45}}}$$

$$= \frac{\sqrt{45}}{16} \left(x^2 - 6x + \frac{23}{3} \right) = \frac{\sqrt{5}}{16} (3x^2 - 18x + 23)$$

báze B je tak

$$B = \left(\overset{\vec{u}_1}{\underset{||}{\frac{1}{2}}}, \overset{\vec{u}_2}{\underset{||}{\frac{\sqrt{3}}{4}(x-3)}}, \overset{\vec{u}_3}{\underset{||}{\frac{\sqrt{5}}{16}(3x^2-18x+23)}} \right)$$

b)

$$\vec{w}_1 = \left\langle \frac{1}{2}, \min x \right\rangle \frac{1}{2} = \left(\int_1^5 \frac{\min x}{2} dx \right) \frac{1}{2} \approx 0,064$$

$$\vec{w}_2 = \left\langle \frac{\sqrt{3}}{4}(x-3), \min x \right\rangle \left(\frac{\sqrt{3}}{4}(x-3) \right) =$$

$$= \left(\int_1^5 \frac{\sqrt{3}}{4}(x-3) \min x dx \right) \left(\frac{\sqrt{3}}{4}(x-3) \right) \approx -0,647(x-3)$$

$$\vec{w}_3 = \left\langle \vec{u}_3, \min x \right\rangle \vec{u}_3 = \left(\int_1^5 \vec{u}_3 \min x dx \right) \vec{u}_3 \approx$$

$$\approx -0,0175(3x^2-18x+23)$$

$$\Rightarrow \vec{w} = \vec{w}_1 + \vec{w}_2 + \vec{w}_3 = \underline{\underline{-0,053x^2 - 0,332x + 1,607}}$$

c)

$$\| \min x - (ax^2 + bx + c) \| \approx \sqrt{\int_1^5 (\min x + 0,053x^2 - 0,332x + 1,601)^2 dx} \approx$$

$$\approx \underline{\underline{0,319}}$$